PAID VU LMS HANDLING by Mam Mehwish 03184148783
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Sure, here's a summary of the MCQs with their options and the correct answers you selected:

1. MCQ: Equivalent form of the exact differential equation: ydx + xdy = 0.
   - a) d(x/y) = 0
   - b) d(xy) = 0
   - c) d(y/x) = 0
   - d) d(x + y) = 0
   - Correct Answer: b) d(xy) = 0

2. MCQ: For f(x,y) = x^2y - xy^2, f(tx,ty) is...
   - a) t^3 f(x, y)
   - b) f(x, y)
   - c) t^2 f(x, y)
   - d) tf(x, y)
   - Correct Answer: c) t^2 f(x, y)

3. MCQ: Which function would satisfy: dy/dx = d^2y/dx^2 = ... = d^n y/dx^n?
   - a) y = Ax^n e^x
   - b) y = Axe^x
   - c) y = Ae^x^2
   - d) y = Ae^x
   - Correct Answer: d) y = Ae^x

4. MCQ: Order of the differential equation: d^2y/dx^2 + 5(dy/dx)^3 - 3y = e^sin x.
   - a) 3

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- b) 0

- c) 1

- d) 2

- Correct Answer: d) 2

5. MCQ: Integrating factor for a non-exact differential equation ydx - (y - 3x - 3) dy = 0.

- a) -y^2

- b) y^2

- c) 1/y^2

- d) -1/y^2

- Correct Answer: c) 1/y^2

6. MCQ: Constant solution of the separable differential equation: dy/dx = e^x+y?

- a) y = 0

- b) y = 1

- c) No constant solution exists

- d) y = a ∈ ℝ

- Correct Answer: a) y = 0

7. MCQ: Integrating factor for the first-order linear differential equation: (x - 1)^3 dy/dx + 4(x - 1)^2 y = cos x.

- a) e^(1/(x-1))

- b) e^(x-1)

- c) (x - 1)^4

- d) ln(x - 1)

- Correct Answer: b) e^(x-1)

8. MCQ: Integrating factor for the non-exact differential equation M(x,y)dx + N(x,y)dy = 0.

- a) 1/(xM - yN), xM - yN ≠ 0

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- b) ∂M/∂y - ∂N/∂x/N

- c) ∂N/∂x - ∂M/∂y/M

- d) 1/(xM + yN), xM + yN ≠ 0

- Correct Answer: d) 1/(xM + yN), xM + yN ≠ 0

9. MCQ: The differential equation dx/dy + x/y = 2 sin y is first-order linear in unknown ___.

- a) variable x

- b) multi-variables x and y

- c) variable y

- d) dy/dx

- Correct Answer: c) variable y

10. MCQ: General solution of the separable differential equation: dy/(1+y^2) = 2xdx.

- a) y = cot^(-1)(x^2 + c)

- b) y = tan(x^2 + c)

- c) y = tan^(-1)(x^2 + c)

- d) y = cot(x^2 + c)

- Correct Answer: c) y = tan^(-1)(x^2 + c)

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Which of the following is an equivalent form of the exact differential equation: ydx + xdy = 0?

$d(\frac{x}{y}) = 0$

$d(xy) = 0$

$d\left(\frac{y}{x}\right) = 0$

$d(x + y) = 0$

For $f(x, y) = x^2y - xy^2$, $f(tx, ty) = \ldots$.

$t^3 f(x, y)$

$f(x, y)$

$t^2 f(x, y)$

$tf(x, y)$

Which of the following function would satisfy: $\frac{dy}{dx} = \frac{d^2y}{dx^2} = \cdots = \frac{d^ny}{dx^n}$?

Select the correct option

- $y = Ax^n e^{x^n}$
- $y = Axe^x$
- $y = Ae^{x^n}$
- $y = Ae^x$

Which of the following function would satisfy: $\frac{dy}{dx} = \frac{d^2y}{dx^2} = \cdots = \frac{d^ny}{dx^n}$?

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- $y = Axe^x$
- $y = Ae^{x^n}$
- $y = Ae^x$

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If the non - exact differential equation M(x,y)dx + N(x,y)dy = 0, is of the form yf(xy)dx + xg(xy)dy = 0, then the integrating factor is - - - -

-
- $\frac{1}{xM - yN}, xM - yN \neq 0$
- $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$
- $\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$
- $\frac{1}{xM + yN}, xM + yN \neq 0$
-

The differential equation $\frac{dx}{dy} + \frac{1}{y}x = 2\sin y$ *is first order linear in unknown* $---$
variable x
multi - variables x and y
variable y
$\frac{dy}{dx}$

General solution of the separable differential equation: $\frac{dy}{1+y^2} = 2xdx$ is---.

Select the correct option

○	$y = \cot^{-1}(x^2 + c)$
○	$y = \tan(x^2 + c)$
○	$y = \tan^{-1}(x^2 + c)$
○	$y = \cot(x^2 + c)$

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